QWEN / THE RESEARCH JOURNEY

Evidence: 19 Oct 2026, 06:14 UTC

Make it work. Measure it. Improve it. Verify it.

29 Sep–19 Oct 2026 • Qwen3-235B-A22B-Instruct-2507-FP8 • SGLang serving

FIXED WORKLOAD	GOOD REQUEST	SCORE / GATE
~4,000 input → 200 output tokens	TTFT ≤ 3,000 ms AND request mean ITL ≤ 20 ms	good requests/s × 200; ≥90% jointly good

A H200: six stages of improvement

16 GPUs • selected milestones, not isolated causal effects

01 Build a trustworthy test

AIPerf → Serving API → Metrics

Slurm smoke test; later a separate PBS cluster.
Pin model, containers, protocol and output checks.
Fix startup races, offline paths and port conflicts.

60 s warmup → 120 s scoring → 30 s grace

02 Repair topology + KV transfer

Prefill 1 · TP4 | Prefill 2 · TP4 → 8-GPU decode

17 named trials: failures are not valid scores.
Request-scoped NIXL enables TP4 → TP8 KV.
Retained: 8 prefill GPUs + 8 decode GPUs.

1,912.300 good tokens/s · 22415 · N48

03 Improve the serving pipeline

Route → Graphs → Transfer

Chunk 4096 + load: 2,140.096 (23114).
Round robin, graphs, MNNVL: 2,592.162.
FlashInfer decode attention: 2,632.610.

N56 pipeline: 2,667.465 · 24263

then

04 Choose kernels for each phase

PREFILL	DECODE
DeepGEMM	auto / HPC-Ops

Prefill and decode have different batch shapes.
DeepGEMM prefill: +1.74% vs fresh control.
Independent repeat supports the H200 gain.

2,716.787 good tokens/s · 24327 · N56

05 Refine chunk size + client load

4096 | 6144 | 8192

Prefill token budget; the prompt stays ~4k.

Chunk 6144 / N56: 2,737.410 (24573).
Chunk 8192 / N58: 2,780.403 (24721).
Auto-decode N60: 2,839.532 (24722).

N60 repeat: 2,836.531 · 24732

06 Combine the useful changes

BEST OBSERVED H200

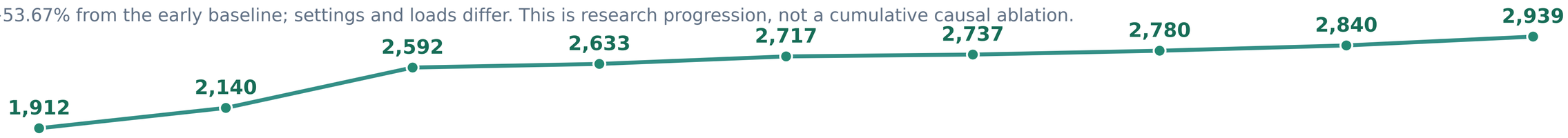
2,938.628 good tokens/s

DeepGEMM prefill · chunk 8192 · HPC-Ops decode
N58 · job 24724 · +5.69% vs fresh N58 control
One accepted deployment; own repeat pending.
Official accuracy and global optimum unverified.

B What the measured progression looks like

good output tokens/s • H200 only

+53.67% from the early baseline; settings and loads differ. This is research progression, not a cumulative causal ablation.



Latest matched N58 change: mean ITL 16.498 → 14.998 ms; TTFT 784.526 → 871.991 ms; joint success 98.907%.

Faster repeated decode intervals can offset slower first-token arrival. This is a plausible explanation, not a profiled causal proof.

C Branches that taught us where to stop

Failure, invalid output and slower performance have different meanings

Larger chunk: 12288
2,714.261 · job 24723
−2.38% at matched N58
CLOSE: slower than chunk 8192

Disable prefill graphs
1,590.313 · job 24426
Only 72.72% jointly good
REJECT: fails the latency gate

Equal TP8 / EP2 layout
Job 24375
Output checks fail
QUARANTINE: score is not valid

Startup / warmup failures
NIXL, router, job 24588
No scored benchmark result
DIAGNOSE: retain logs, no score

D H100: a separate research lane

PBS • 16 GPUs • no pooled H100/H200 ranking

Re-measure on the second platform

H200 DeepGEMM gains did not transfer reliably.
Faster weight loading did not prove goodput gain.
Both-phase HPC-Ops earned repeated support.

N50 repeat: 2,371.341 · job 221339

Refine the operating load

N52 repeat: 2,411.304 · job 221589

2,430.891 good tokens/s

N54 · job 221590 · best observed, own repeat pending

Profile → test communication

Ring all-reduce appears in sealed GPU traces.
N54 repeat → N56 → prefill NCCL Tree trial.
221645 → 221646 → 221669; outcomes pending.

Prove actual Tree activation before crediting it.

E What makes the next improvement credible

Planned / pending at the evidence cutoff; not new measurements

1 Reproduce + bound the load

Fresh control 24736; interaction repeat 24737.
N60 interaction 24738; auto N62 trial 24733.
Promote only if the gain survives local gates.

2 Establish official accuracy

Freeze model, payloads, scripts and runtime.
Run the organiser harness and reference tasks.
≥99% reference score is still unverified.

3 Profile, then choose the next target

Profile the actual winning H200 combination.
Test demonstrated kernel / collective costs.
Close losing branches; avoid blind larger sweeps.

THE LOOP USED THROUGHOUT

Inspect logs → Hypothesis → Freeze → Measure → Validate → Repeat → Promote / close

Preserve raw logs + failed runs + frozen configs + checksums + controls + credit ledger + dated decisions.

READING KEY Arrows = research progression. N = active requests. All throughput figures = good output tokens/s.

Sources: REPORT.md, evidence.json and the linked run reviews. Snapshot only; no live queue status, official accuracy or global-optimum claim.